

Moksh Buddhadev

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EDUCATION

Manipal University Jaipur

2023 – 2027 (Expected)

B.Tech in Computer Science and Engineering – CGPA: 8.39/10

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Deep Learning, NLP, Computer Vision, DBMS, Operating Systems, Computer Networks

PROJECTS

- Credit Risk Assessment – *Predictive Creditworthiness Analysis System*
 - Developed a leakage-free ML pipeline on 30,000 records, benchmarking 17 classifiers (CatBoost, LightGBM, XGBoost, Stacking Ensemble, regularised DNN); achieved state-of-the-art AUC of **0.7730**, surpassing prior benchmark by 0.003.
 - Boosted minority-class Sensitivity by **31%** and F1-Score by **9%** via SMOTE-based resampling (post-scaling), threshold optimisation via validation-set grid search, and Bayesian hyperparameter tuning (Optuna, CV-AUC = 0.9404).
 - Engineered 9 domain-informed features (MAX_DELAY, DELAY_COUNT, UTIL_RATE, PAY_RATIO) validated by SHAP TreeExplainer, accounting for 24% of total Random Forest importance; threshold-optimised LightGBM achieved F1 = 0.5318, G-Mean = 0.6801.
 - Applied rigorous model explainability techniques (SHAP) to communicate feature contributions to stakeholders, bridging the gap between technical outputs and business decision-making.
- ReguLens-AI – *Regulatory Document Intelligence System*
 - Architected an end-to-end Retrieval-Augmented Generation (RAG) pipeline for financial/regulatory QA, integrating Groq LLM (Llama 3) with FAISS vector database and sentence-transformer embeddings for sub-second semantic retrieval.
 - Built modular FastAPI backend with decoupled ingestion (PyMuPDF), chunking, embedding, retrieval, prompt engineering, and LLM inference layers; Streamlit frontend supports real-time NLP querying over RBI, KYC, and compliance corpora.
 - Implemented session-aware REST API with scalable multi-document indexing; eliminated manual document review bottlenecks for dense regulatory texts and improved query throughput significantly.
 - Created prompt engineering strategies to reduce hallucinations and improve factual grounding, ensuring reliable responses over high-stakes compliance documents.
- FloatChat AI – *ARGO Oceanographic Data Intelligence System*
 - Prepared NLP query engine integrating Groq Llama 3.1 (70B) with RAG, achieving **95% query success rate** and sub-3s end-to-end response latency; hybrid SQL + tool-based routing with PostGIS spatial intelligence enabled advanced geospatial analysis.
 - Reduced query failure rate by **40%** via session-aware query handling; built scalable FastAPI backend and Streamlit dashboard enabling domain experts to extract insights through natural language without SQL expertise.
 - Engineered a structured tool-calling architecture that dynamically routes user queries to SQL execution or geospatial APIs, supporting complex analytical workflows over oceanographic float datasets.

TECHNICAL SKILLS

Languages	Python, C++, Java, SQL
ML / AI	Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow/Keras, SHAP, Optuna
LLM & GenAI	RAG, LLM Integration (Groq, Llama 3), Prompt Engineering, FAISS, Sentence Transformers
Backend & Tools	FastAPI, PostgreSQL, PostGIS, PyMuPDF, Git, REST APIs, Streamlit
Specialisations	Feature Engineering, Class Imbalance Handling, Hyperparameter Tuning, Model Explainability

LEADERSHIP & ORGANIZATIONS

- ACM, Manipal University Jaipur
- 2024 – 2025
- Team Head – Finance
- Managed budgeting, financial planning, and sponsorship strategy for AI/ML workshops and technical events; ensured institutional compliance and optimised cost efficiency across 200+ participant initiatives.
 - Coordinated cross-functional communication between technical teams and sponsors, strengthening organisational partnerships and enhancing event execution quality.